"A Robust Image Steganography Framework Using Edge Detection and CNN for Secure Data Hiding"

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Table of Contents

- Abstract
- Introduction
- Problem Statement
- Objectives
- Applications
- Literature Survey & Research Gaps Identified
- Proposed System Design
- Dataset Description
- Software and Hardware Components
- Performance Evaluation
- Conclusion
- References

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Abstract

Steganography has emerged as an effective technique for secure communication, where secret data is concealed within digital images to prevent unauthorized access. However, conventional methods such as Least Significant Bit (LSB) substitution suffer from low robustness, reduced security, and high vulnerability to detection through steganalysis. These limitations make traditional steganography unsuitable for modern secure communication, especially when images undergo common processing operations such as compression, noise addition, and filtering.

To overcome these challenges, this project proposes a robust image steganography framework that integrates edge detection with Convolutional Neural Networks (CNN). Edge detection is applied to the cover image to identify high-frequency regions where embedding is less noticeable to the Human Visual System (HVS). A CNN model is then used to optimize both embedding and extraction by learning suitable pixel positions and minimizing distortion. Experimental evaluation shows high PSNR and SSIM values for maintaining visual quality, along with low BER for accurate data recovery, confirming that the proposed approach achieves a strong balance between imperceptibility, robustness, and security.

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Introduction

In the modern digital era, secure communication has become a major concern due to the rapid growth of internet usage and the frequent sharing of multimedia data. Images are one of the most commonly exchanged forms of digital content through social media, cloud platforms, and communication networks, making them an ideal medium for hiding confidential information. Unlike encryption, which makes the presence of secret data obvious, steganography hides information inside images in such a way that it remains invisible to unauthorized users.

Traditional image steganography techniques such as Least Significant Bit (LSB) embedding are widely used because they are simple and provide good visual quality. However, these methods are weak in terms of robustness and security, as they can be easily detected using steganalysis and the hidden data may be destroyed when the image undergoes compression, noise, or filtering. To overcome these limitations, this project proposes a robust image steganography framework that combines edge detection and Convolutional Neural Networks (CNN). Edge detection helps identify high-frequency regions where embedding is less noticeable, while CNN improves the embedding and extraction process by learning optimal pixel locations, resulting in improved imperceptibility, robustness, and secure data recovery.

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Problem Statement

- Traditional image steganography techniques such as **LSB embedding** do not provide a proper balance between **security, imperceptibility, and robustness**.
- These methods are easily detectable through **steganalysis**, making hidden communication insecure.
- The hidden data can be lost or corrupted when the stego image undergoes common image processing operations such as **JPEG compression, noise addition, and filtering**.
- Uniform embedding across the image may create **noticeable distortion**, reducing the visual quality of the stego image.
- Hence, there is a need for a **robust and intelligent steganography framework** that uses **edge detection** to select suitable embedding regions and **CNN** to optimize embedding and extraction while maintaining high image quality and strong attack resistance.

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Objectives

1. The Primary objective is To design and develop a **robust image steganography framework** for secure data hiding in digital images, ensuring that the secret information remains confidential and the stego image appears visually similar to the original cover image..
2. To apply **edge detection techniques** such as **Canny, Sobel, and HED (Holistically Nested Edge Detection)** to identify **high-frequency edge regions**, because embedding in these regions is less noticeable to the **Human Visual System (HVS)** and improves imperceptibility.
3. To integrate a **CNN-based encoder–decoder model** to optimize the **embedding and extraction process**, by automatically learning suitable pixel positions, reducing embedding distortion, and improving the accuracy of recovering the hidden message from the stego image.
4. To improve the system's **imperceptibility and visual quality** by minimizing visible distortion in the stego image and achieving higher performance in terms of **PSNR (Peak Signal-to-Noise Ratio)** and **SSIM (Structural Similarity Index)** compared to traditional steganography methods.
5. To ensure strong **robustness and security** by making the hidden data resistant to common image processing operations such as **JPEG compression, noise addition, and filtering**, while maintaining **low BER (Bit Error Rate)** and improving resistance against **steganalysis detection techniques**..

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Applications

•**Secure Communication**
Hiding confidential messages within images for safe transmission over public networks.
•**Military & Defense**
Used for covert communication and secure exchange of sensitive information.
•**Banking & Financial Security**
Protecting transaction details, credentials, and sensitive financial data.
•**Data Privacy & Protection**
Safeguarding personal data such as medical records, legal documents, and identity information.
•**Digital Watermarking & Copyright Protection**
Embedding ownership information in images to prevent unauthorized use.
•**Cybersecurity Applications**
Preventing data leakage by hiding information in seemingly harmless files.
• **Secure Cloud Storage**
Storing sensitive data inside images before uploading to cloud platforms.
•**Confidential Document Transmission**
Sending classified information without attracting attention.

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Literature Survey

S. No	Year	Authors	Title	Outcomes
1	2025	Al-Rawashdeh, Rahman & Niazi	Robust Image Steganography Approach Based on Edge Detection Combined with CNN Algorithm	Improved imperceptibility and robustness using edge regions + CNN based embedding and extraction.
2	2021	Ray, Banerjee & Bandyopadhyay	Image Steganography Using Deep Learning Based Edge Detection	Edge-aware deep learning improves secure embedding and reduces detectability
3	2020	Liu, Tang & Luo	Secure and Robust Image Steganography Based on Convolutional Neural Networks	CNN improves embedding optimization and accurate recovery of hidden data.
4	1986	Canny	A Computational Approach to Edge Detection	Introduced Canny edge detection, useful for identifying high-frequency regions for hiding data.
5	2018	Hussain et al.	A Survey of Image Steganography Techniques	Highlights that LSB methods are simple but weak against compression, noise, and steganalysis.

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S. No	Year	Authors	Title	Outcomes
6	2025	Zhang, Ji & Lv	Edge-Guided Dual-Stream U-Net for Secure Image Steganography	Deep U-Net based edge-guided model gives high-quality stego images and accurate extraction.
7	2023	Sultana et al.	Hybrid Edge Detection and LBP Code-Based Robust Image Steganography Method	Combines edge + texture features to improve security and imperceptibility.
8	2025	Almuhammadi et al.	Secure Edge-Guided Adaptive Image Steganography Using HED-Based Attention Maps and CNN	HED attention maps + CNN improve adaptive embedding and robustness.
9	2024	Habiban,Hamad & Mohsin	Hybrid Edge Detection Methods in Image Steganography for High Embedding Capacity	Hybrid edge detection improves embedding capacity while maintaining good PSNR/SSIM.
10	2025	Singh & Singh	Improved Edge-Based Image Steganography Using Segmentation and Random LSB Substitution	Random edge embedding improves security compared to standard LSB.

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Research Gaps

1. **Traditional LSB-based Steganography** is simple but it is weak against steganalysis and fails when the image undergoes compression, noise addition, or filtering.
2. Many existing steganography methods embed data **uniformly across the image** without selecting the best regions, which increases distortion and makes hidden data easier to detect.
3. Even though **edge-based embedding** improves imperceptibility, classical edge detection methods are **rule-based** and do not adapt well to different image types, textures, and complexities.
4. Several deep learning (CNN) based approaches improve embedding and extraction, but **without edge guidance**, CNN models may still cause distortion in smooth regions and reduce invisibility.
5. The biggest research gap remains achieving an effective balance between:

   Payload Capacity

   Imperceptibility (PSNR, SSIM)

   Robustness (BER under attacks)

   Security (resistance to steganalysis)
6. Most existing methods do not support adaptive embedding strength based on image complexity, which affects capacity and invisibility.

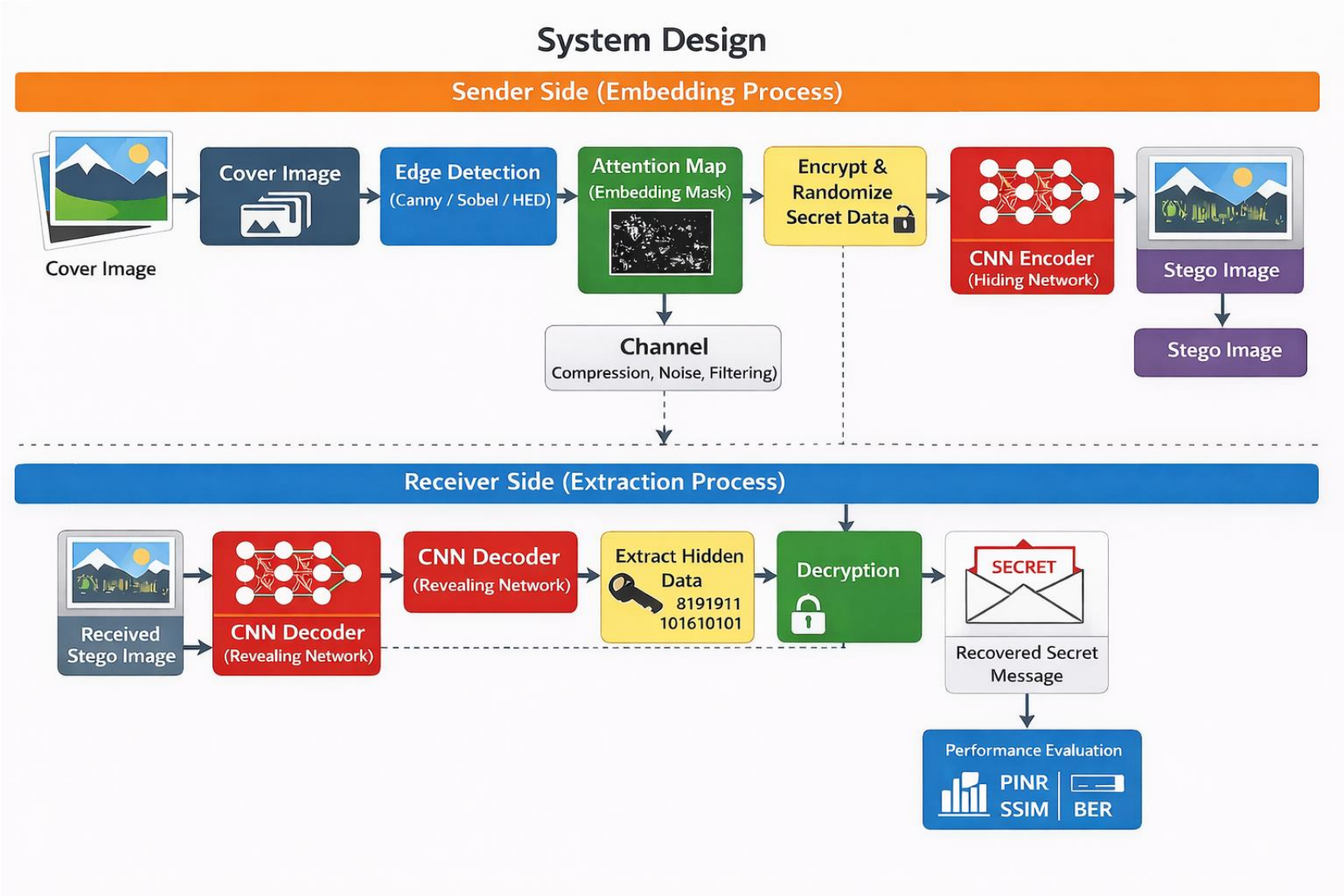
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Proposed System

- The proposed system develops a **robust image steganography framework** using **Edge Detection and CNN** for secure data hiding.
- The cover image is processed using **edge detection techniques** (Canny / Sobel / HED) to identify **high-frequency edge regions** where embedding is less noticeable to the Human Visual System (HVS).
- An **edge-based attention map (embedding mask)** is generated to select suitable embedding locations.
- The secret message is **encrypted and randomized** before embedding to improve security.
- A **CNN Encoder (Hiding Network)** adaptively embeds the secret data mainly into edge regions while minimizing distortion and maintaining image quality.
- The output produced is a **stego image** that looks almost identical to the original cover image.
- At the receiver side, a **CNN Decoder (Revealing Network)** extracts the hidden message accurately from the stego image.
- The system is evaluated using **PSNR, SSIM, and BER** to ensure high imperceptibility, robustness, and reliable extraction even after compression, noise, and filtering attacks.

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System Design



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Dataset Description

- To evaluate the proposed **Edge Detection + CNN based Image Steganography Framework**, standard benchmark image datasets are used to test the system under different image types, textures, and resolutions. The main datasets considered are:

- **BOSS Base Dataset:**
  Contains **10,000 grayscale images** with a resolution of **512×512**. It is widely used as a benchmark dataset in image steganography research for evaluating imperceptibility and steganalysis resistance.

- **USC-SIPI Dataset:**
  A standard digitized image dataset containing images of different resolutions such as **256×256, 512×512, and 1024×1024**. It is useful for testing the robustness of the steganography model across different image qualities and structures.

- **Tiny ImageNet Dataset:**
  Contains **100,000 color images** of resolution **64×64** across **200 classes**. This dataset provides diverse image content and helps evaluate the generalization ability of the CNN model.

These datasets support effective evaluation of the proposed system using performance metrics such as **PSNR, SSIM, and BER**, ensuring the framework performs well across different real-world image scenarios.

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Software and Hardware Requirements

SOFTWARE REQUIREMENS:

- **Operating System:** Windows 10/11 or Linux
- **Programming Language:** Python
- **Development Tools:** Jupyter Notebook / VS Code / Google Colab
- **Libraries / Frameworks:**

  OpenCV (Edge Detection: Canny, Sobel, etc.)

  NumPy (Image processing operations)

  TensorFlow / PyTorch (CNN model implementation)

  Matplotlib (Graphs and evaluation visualization)

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HARDWARE REQUIREMENS:

- **Processor (CPU):** Intel i5 / i7 or AMD Ryzen 5 / Ryzen 7
- **RAM:** Minimum 8GB (Recommended 16GB)
- **Storage:**

  Minimum 20GB free space

  SSD recommended for faster dataset loading and training
- **System Type:** 64-bit system
- **Internet Connection:** Required for dataset download and library installation
- **Input Devices:** Keyboard and Mouse (for development and training workflow)

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Performance Evaluation

- **PSNR (Peak Signal-to-Noise Ratio)**

Measures image quality after embedding; higher PSNR indicates better imperceptibility.

- **SSIM (Structural Similarity Index)**

Evaluates structural similarity between cover and stego images; values closer to 1 indicate high s

- **MSE (Mean Squared Error)**

Measures pixel-level difference; lower MSE means less distortion in the stego image.

- **BER (Bit Error Rate)**

Indicates accuracy of extracted data; lower BER ensures reliable message recovery.

- **Visual Quality Analysis**

Ensures stego image appears identical to original (no visible changes).

- **Robustness Testing**

Evaluated under noise, compression, and filtering to test system reliability.

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Conclusion

The proposed system for ***Robust Image Steganography Using Edge Detection and CNN*** provides an efficient and secure method for hiding confidential data within digital images. By utilizing edge detection techniques, the system identifies suitable regions for embedding, ensuring that the visual quality of the image remains unaffected. The integration of a CNN-based encoder–decoder model enables intelligent and adaptive data embedding and extraction.

The performance evaluation demonstrates that the system achieves high imperceptibility, strong robustness, and accurate data recovery. Metrics such as PSNR, SSIM, and BER confirm that the stego images maintain quality while securely preserving hidden information. Overall, the approach offers a reliable solution for secure communication and data protection in modern digital environments.

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Thank You

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